

LOKESHWARAN P

EMBEDDED & BACKEND ENGINEER

CONTACT

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Alternate Link
- github.com/Lokesh-Spectre

SKILLS

- General
- AI Power user
 - Linux Enthusiast
- Languages
- C/C++
 - Python
 - JavaScript
 - Dart

Frameworks/SDKs

- ESP-IDF
- Flutter
- NestJS
- PySide6

PROJECTS

SCAN FORGE
Generates a 3D Object file by scanning real world object.
It contains a embedded firmware, android app and a desktop app

NEO GRID
A cost effective solution for IOT based Line-Breakage-Detection-System for overhead power lines.

COSMO STORE
A context based document retrieval system for high throughput environments, designed for opensource LLMs for data privacy



PROFILE

Engineering student specializing in embedded firmware and backend development. Experienced in production-level ESP32-S3 firmware using ESP-IDF and FreeRTOS, and in building distributed backends with NestJS and Docker. Seeking an internship to apply full-stack system design and development expertise.



WORK EXPERIENCE

EMBLOCK TECH 2023 - 2025
Chief Product Officer

- Developed and maintained backend systems for multiple projects, including two e-commerce platforms and a POS billing system, using NestJS, PostgreSQL, and Docker.
- Independently developed hardware and firmware for an ESPressif's MCU based IoT device using ESP-IDF, FreeRTOS, and MQTT communication with OTA updates.
- Designed and implemented REST APIs, authentication workflows, and database schemas optimized for reliability and scalability



LEADERSHIP

COLLEGE CLUB

- Current Technical Lead of CloudOps Club VIT. We've conducted 1 hackathon and 2 workshops among other events.

WORK

- Led two frontend teams and one embedded systems team at Emblock Tech, overseeing projects from inception to successful completion.



EDUCATION

Computer Science Engineering (AI/ML) 2023-2027
B.Tech
Vellore Institute of Technology Chennai



LANGUAGES

- English (Fluent)
- Tamil (Fluent)